

SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

VAYU-SARTHI

Plans the air tasking day, and re-plans it in seconds when something changes, moving as few aircraft as possible.

- **Problem Statement ID** – 26250
- **Problem Statement Title** – Air Power: Dynamic Air Operations & Resource Optimisation
- **Theme** – Transportation & Logistics
- **PS Category** – Software
- **Team ID** –
- **Team Name** – NexaBuild



VAYU-SARTHI

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

1 Detailed explanation of the proposed solution

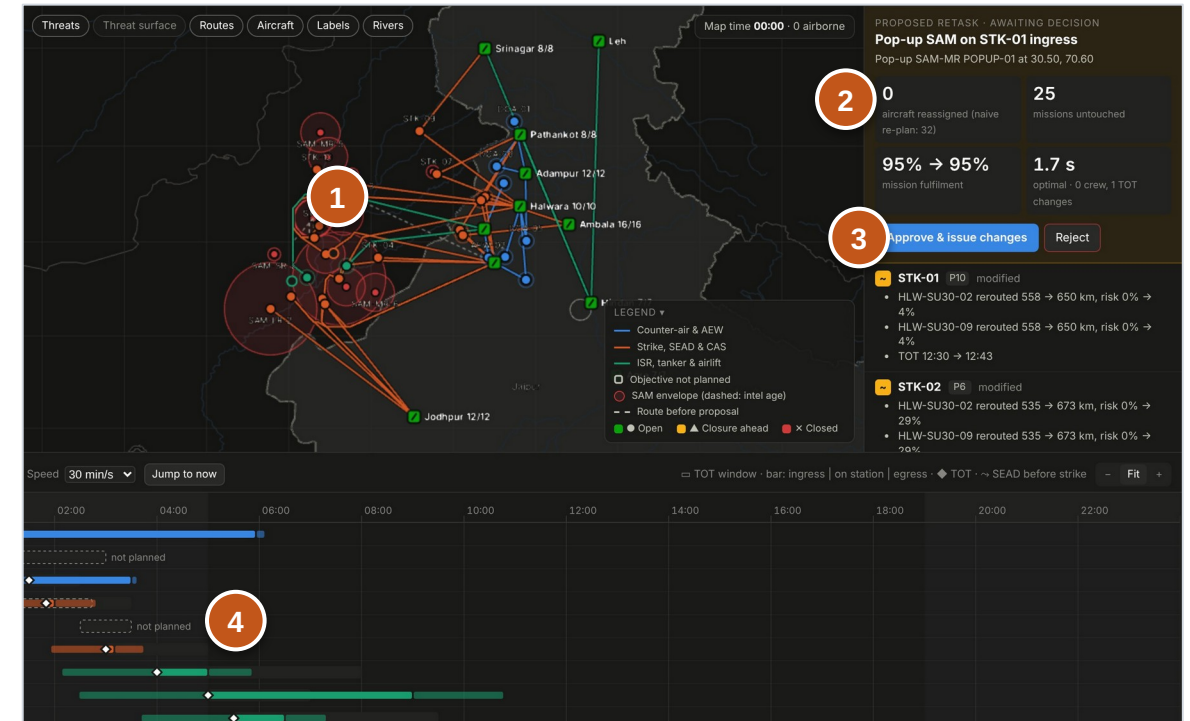
- One live picture: aircraft, crews, weapons, weather, threats
- Builds the full air tasking plan for the day in under 10 s
- When something changes, proposes the **smallest fix** as a diff
- The commander approves or rejects; nothing moves on its own

2 How it addresses the problem

The problem asks for	What VAYU-SARTHI does
Dynamic retasking	Re-plans in ~3 s; the rest of the plan stays put
Resource optimisation	One model: aircraft, crew rest, weapons, tankers, reserve
Weather, uncertainty	Fog forecast from real airport data; 2,000 simulated days
Decision support	Gives a reason for each choice; a human approves it

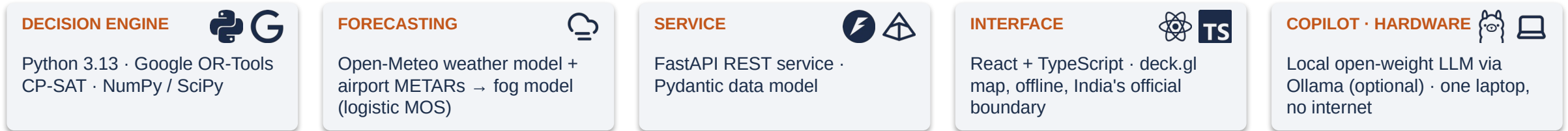
3 Innovation and uniqueness of the solution

- Changes as little as possible: a diff, not a brand-new plan
- Says what it would take to fit any mission it had to drop
- Plain-language copilot that answers only from the engine
- Runs offline on one laptop; also plans flood and earthquake relief

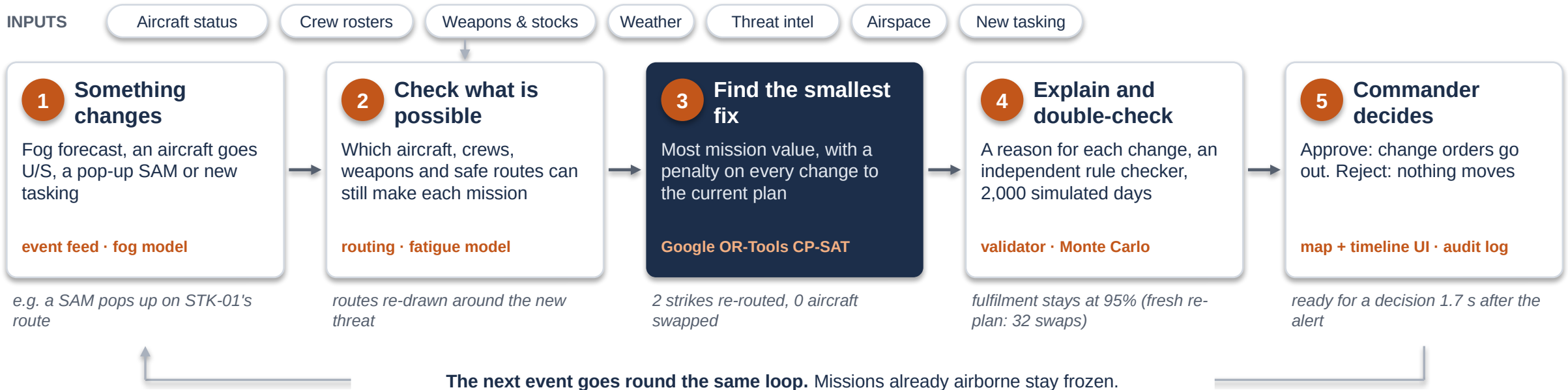


- 1 A SAM pops up on STK-01's route; old routes show dashed
- 2 Proposal: 2 strikes re-routed, 0 aircraft swapped (a fresh re-plan swaps 32)
- 3 The commander approves or rejects, 1.7 s after the alert
- 4 Synchronisation matrix: every sortie's timing, before and after

❖ Technologies to be used (e.g. programming languages, frameworks, hardware)



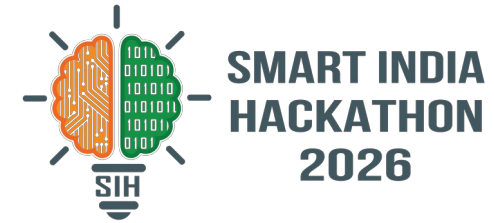
❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



Built and working: 55 automated engine tests · a browser test that clicks through the whole flow · every plan passes the independent rule checker

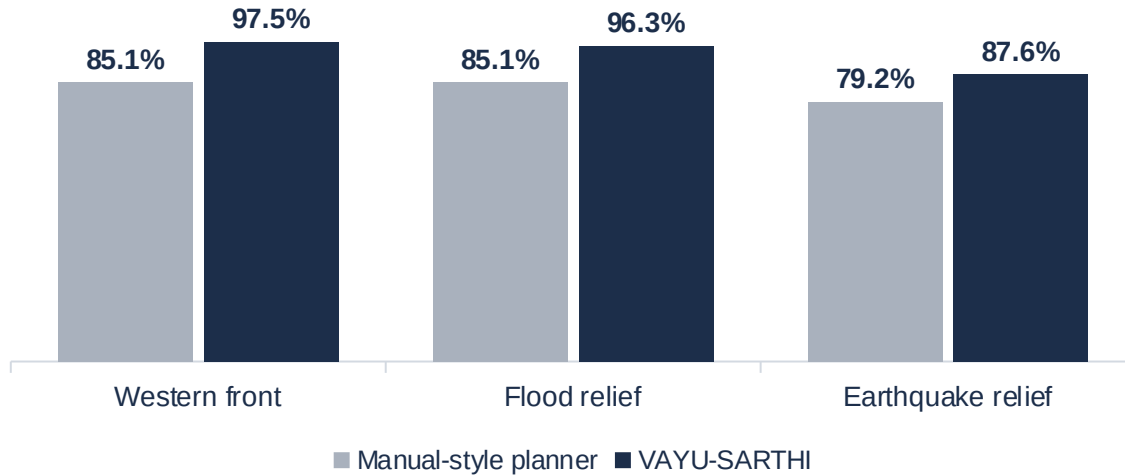


FEASIBILITY AND VIABILITY



❖ Analysis of the feasibility of the idea

Mission fulfilment, same fleet and rules (20 scenarios each)



2.8 s

to re-plan when the busiest base fogs in

-71%

aircraft moved vs a fresh re-plan (14.5 vs 49.5)

70%

of fog hours forecast (raw weather model: 26%)

- **Already built and tested:** engine, map, timeline, copilot; 55 tests
- Every plan passes an independent rule checker (crew rest, stocks, weather, airspace)
- Open-source stack on one laptop, offline: **no licence or cloud cost**

❖ Potential challenges and risks

Real data is classified; legacy systems such as IACCS

Commanders may not trust machine advice

Theatre scale: 100+ missions a day

Forecasts drift with season and region

Network cut off (degraded links)

❖ Strategies for overcoming these challenges

One adapter per data source behind one world model; notional data until access is granted

A reason for every choice, a human approves every change, an independent checker (DRDO ETAI principles)

~40 missions plan in 4–10 s today. Next: large-neighbourhood search, split by sector and time

Probabilities, not yes/no; tested on a held-out winter; retrained per region and season

Runs fully offline today. Next: a node at each base that syncs when the link returns

❖ Potential impact on the target audience



Air tasking cell, Air HQ and Commands

Seconds, not hours

to re-plan after fog, an aircraft going U/S or a pop-up threat



Squadrons and base operations

-71% churn

14.5 aircraft moved instead of 49.5 when a base fogs in



Aircrew

Rested crews

rest, fatigue and night currency are hard rules in every plan



NDMA and state disaster response

+12.9 pts relief

of requested flood-relief tonnage lifted by the same fleet

❖ Benefits of the solution (social, economic, environmental, etc.)

Operational

+12.4 pts mission fulfilment from the same aircraft and crews

Economic

Open-source, one laptop, no licences; idle aircraft become ground spares at no cost

Social

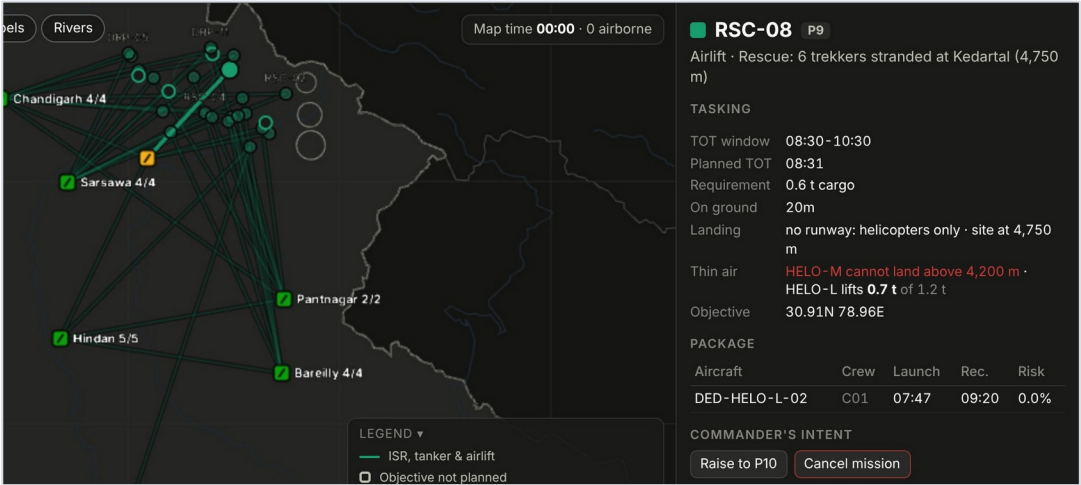
Flood and earthquake relief planned rescues-first: 94.5% and 88.1% of relief lifted

Environmental

Every sortie carries a cost in the plan, so none is flown without need

Strategic

Indigenous and air-gapped: no foreign cloud, model or map service



The screenshot displays a mission planning interface. On the left, a map of India shows various locations marked with green checkmarks and labels: Chandigarh 4/4, Sarsawa 4/4, Hindan 5/5, Pantnagar 2/2, and Bareilly 4/4. A legend at the bottom indicates 'ISR, tanker & airlift' and 'Objective not planned'. On the right, a detailed tasking panel for 'RSC-08 P9' is shown. It includes mission details such as 'Airlift - Rescue: 6 trekkers stranded at Kedartal (4,750 m)', 'TOT window 08:30-10:30', 'Planned TOT 08:31', 'Requirement 0.6 t cargo', 'On ground 20m', 'Landing no runway: helicopters only - site at 4,750 m', 'Thin air HELO-M cannot land above 4,200 m - HELO-L lifts 0.7 t of 1.2 t', and 'Objective 30.91N 78.96E'. A 'PACKAGE' table lists aircraft, crew, launch, rec., and risk. The 'COMMANDER'S INTENT' section shows 'Raise to P10' and 'Cancel mission' buttons.

Aircraft	Crew	Launch	Rec.	Risk
DED-HELO-L-02	C01	07:47	09:20	0.0%

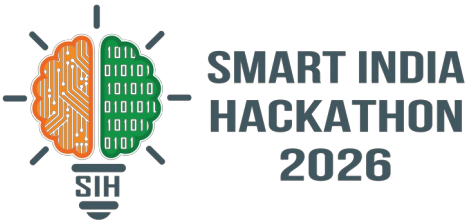
Same engine, earthquake relief: thin air at 4,750 m rules out the medium helicopter

@SIH Idea submission- Template

5



RESEARCH AND REFERENCES



❖ Details / Links of the reference and research work

Reference	Link	What we took from it
1 IAF AI decision-support tools for IACCS (IDRW)	idr.wa.gov/?p=383033	Where a tool like this would plug in
2 DRDO ETAI framework for trustworthy AI in defence (2024)	indiaai.gov.in	Human approval, reasons, independent checks
3 Kessel Run KRADOS and Slapshot: ATO in the cloud (USAF)	airand.spaceforces.com	What exists abroad; the gap is minimal-change retasking
4 DARPA Adapting Cross-Domain Kill-Webs (ACK)	darpa.mil	A decision aid to task and retask quickly
5 Google OR-Tools, CP-SAT constraint solver	developers.google.com/optimization	The planning and retasking engine
6 Glahn & Lowry (1972), Model Output Statistics, J. Appl. Meteor. 11	doi.org/10.1175/1520-0450(1972)011	How the fog forecast is built
7 Borbély (1982), two-process model of sleep; SAFTE in ICAO FRMS	icao.int (FRMS modelling)	Crew fatigue and rest windows
8 Minimal-deviation rescheduling under disruption (METU)	avesis.metu.edu.tr	A penalty on changing the current plan
9 Open-Meteo forecasts · IEM METAR archive · Natural Earth (India view)	open-meteo.com · mesonet.agron.iastate.edu	Real weather, airport observations, maps
10 Flexible Use of Airspace in India (ICAO APAC IP08, 2023)	icao.int	Airspace that opens and closes in the day

Our prototype (open source): github.com/DeepakSinghhh/Air-power-Dynamic · all scenario data is notional; maps use India's official boundary